

Solucion tarea 8

1. Preparativos

Subimos el fichero cite75_99.txt a Hadoop dentro de la carpeta /practicass

```
(kali㉿nodo1)-[~/Downloads]
$ /home/kali/Desktop/hadoop/bin/hdfs dfs -mkdir -p /practicass

(kali㉿nodo1)-[~/Downloads]
$ /home/kali/Desktop/hadoop/bin/hdfs dfs -put cite75_99.txt /practicass
```

2. Archivo Java

Carpeta donde está MyJob.java

```
(kali㉿nodo1)-[~/Desktop/practica8]
$ ls
cite75_99.txt  'MyJob$MapClass.class'  'MyJob$Reduce.class'  MyJob.class  MyJob.jar  MyJob.java

(kali㉿nodo1)-[~/Desktop/practica8]
$
```

Exportar librerías de java

```
(kali㉿nodo1)-[~/Desktop/practica8]
$ export HADOOP_CLASSPATH=$JAVA_HOME/lib/tools.jar
```

Compilamos

```
(kali㉿nodo1)-[~/Desktop/practica8]
$ /home/kali/Desktop/hadoop/bin/hadoop com.sun.tools.javac.Main MyJob.java
Note: MyJob.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
```

Creamos el JAR

```
(kali㉿nodo1)-[~/Desktop/practica8]
$ jar cvf MyJob.jar MyJob*.class

added manifest
adding: MyJob$MapClass.class(in = 1575) (out= 631)(deflated 59%)
adding: MyJob$Reduce.class(in = 2192) (out= 933)(deflated 57%)
adding: MyJob.class(in = 1977) (out= 932)(deflated 52%)

(kali㉿nodo1)-[~/Desktop/practica8]
$
```

Ejecutamos el Job

```
(kali@nodo1)-[~/Desktop/practica8]
$ /home/kali/Desktop/hadoop/bin/hadoop jar MyJob.jar MyJob \
/practicass/cite75_99.txt /resultado7

2026-02-02 09:11:35,328 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at nodo1/192.168.1.170:8032
2026-02-02 09:11:36,161 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/kali/.staging/job_1770036565126_0001
2026-02-02 09:11:36,828 INFO input.FileInputFormat: Total input files to process : 1
2026-02-02 09:11:37,563 INFO mapreduce.JobSubmitter: number of splits:2
2026-02-02 09:11:37,966 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1770036565126_0001
2026-02-02 09:11:37,966 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-02-02 09:11:38,248 INFO conf.Configuration: resource-types.xml not found
2026-02-02 09:11:38,248 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-02-02 09:11:38,714 INFO impl.YarnClientImpl: Submitted application application_1770036565126_0001
2026-02-02 09:11:38,960 INFO mapreduce.Job: The url to track the job: http://nodo1:8088/proxy/application_1770036565126_0001/
2026-02-02 09:11:38,961 INFO mapreduce.Job: Running job: job_1770036565126_0001
2026-02-02 09:11:52,445 INFO mapreduce.Job: Job job_1770036565126_0001 running in uber mode : false
2026-02-02 09:11:52,446 INFO mapreduce.Job: map 0% reduce 0%
2026-02-02 09:12:17,782 INFO mapreduce.Job: map 13% reduce 0%
2026-02-02 09:12:20,084 INFO mapreduce.Job: map 26% reduce 0%
2026-02-02 09:12:23,688 INFO mapreduce.Job: map 30% reduce 0%
2026-02-02 09:12:27,076 INFO mapreduce.Job: map 36% reduce 0%
2026-02-02 09:12:29,667 INFO mapreduce.Job: map 41% reduce 0%
2026-02-02 09:12:32,576 INFO mapreduce.Job: map 45% reduce 0%
2026-02-02 09:12:35,898 INFO mapreduce.Job: map 55% reduce 0%
2026-02-02 09:12:38,721 INFO mapreduce.Job: map 64% reduce 0%
2026-02-02 09:12:43,471 INFO mapreduce.Job: map 65% reduce 0%
2026-02-02 09:12:44,689 INFO mapreduce.Job: map 67% reduce 0%
2026-02-02 09:12:48,133 INFO mapreduce.Job: map 75% reduce 0%
2026-02-02 09:12:51,477 INFO mapreduce.Job: map 85% reduce 0%
2026-02-02 09:12:52,543 INFO mapreduce.Job: map 93% reduce 0%
2026-02-02 09:12:53,553 INFO mapreduce.Job: map 100% reduce 0%
2026-02-02 09:13:11,641 INFO mapreduce.Job: map 100% reduce 77%
2026-02-02 09:13:17,903 INFO mapreduce.Job: map 100% reduce 88%
2026-02-02 09:13:24,037 INFO mapreduce.Job: map 100% reduce 99%
2026-02-02 09:13:25,053 INFO mapreduce.Job: map 100% reduce 100%
2026-02-02 09:13:26,070 INFO mapreduce.Job: Job job_1770036565126_0001 completed successfully
2026-02-02 09:13:26,176 INFO mapreduce.Job: Counters: 54
    File System Counters
        FILE: Number of bytes read=594240678
        FILE: Number of bytes written=892190336
        FILE: Number of read operations=0
        FILE: Number of large read operations=0
        FILE: Number of write operations=0
        HDFS: Number of bytes read=264079739
        HDFS: Number of bytes written=158078539
        HDFS: Number of read operations=11
        HDFS: Number of large read operations=0
        HDFS: Number of write operations=2
        HDFS: Number of bytes read erasure-coded=0
    Job Counters
        Launched map tasks=2
        Launched reduce tasks=1
```

Vemos el resultado

```
(kali@nodo1)-[~/Desktop/practica8]
$ /home/kali/Desktop/hadoop/bin/hdfs dfs -cat /resultado7/part-r-00000 | head

"CITED" "CITING"
1      3964859,4647229
10000  4539112
100000 5031388
1000006 4714284
1000007 4766693
1000011 5033339
1000017 3908629
1000026 4043055
1000033 4190903,4975983
cat: Unable to write to output stream.

(kali@nodo1)-[~/Desktop/practica8]
$
```

3. Ejecutar Hadoop Streaming con Shell

```
(kali@nodo1)-[~/Desktop/practica8]
$ /home/kali/Desktop/hadoop/bin/hadoop jar \
/home/kali/Desktop/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
-input /practicas/cite75_99.txt \
-output /resultado8 \
-mapper "cut -f 2 -d ." \
-reducer "uniq"

packageJobJar: [/tmp/hadoop-unjar10558454949675590892/] [] /tmp/streamjob10981372741324479621.jar tmpDir=null
2026-02-02 09:16:56,612 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at nod01/192.168.1.170:8032
2026-02-02 09:16:56,829 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at nod01/192.168.1.170:8032
2026-02-02 09:16:57,180 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/kali/.staging/job_1770036565126_0002
2026-02-02 09:16:57,606 INFO mapred.FileInputFormat: Total input files to process : 1
2026-02-02 09:16:57,633 INFO net.NetworkTopology: Adding a new node: /default-rack/192.168.1.170:9866
2026-02-02 09:16:57,636 INFO net.NetworkTopology: Adding a new node: /default-rack/192.168.1.171:9866
2026-02-02 09:16:57,891 INFO mapreduce.JobSubmitter: number of splits:2
2026-02-02 09:16:58,207 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1770036565126_0002
2026-02-02 09:16:58,207 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-02-02 09:16:58,421 INFO conf.Configuration: resource-types.xml not found
2026-02-02 09:16:58,422 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-02-02 09:16:58,503 INFO impl.YarnClientImpl: Submitted application application_1770036565126_0002
2026-02-02 09:16:58,544 INFO mapreduce.Job: The url to track the job: http://nod01:8088/proxy/application_1770036565126_0002/
2026-02-02 09:16:58,546 INFO mapreduce.Job: Running job: job_1770036565126_0002
2026-02-02 09:17:08,244 INFO mapreduce.Job: Job job_1770036565126_0002 running in uber mode : false
2026-02-02 09:17:08,246 INFO mapreduce.Job: map 0% reduce 0%
2026-02-02 09:17:29,917 INFO mapreduce.Job: map 16% reduce 0%
2026-02-02 09:17:34,060 INFO mapreduce.Job: map 33% reduce 0%
2026-02-02 09:17:44,169 INFO mapreduce.Job: map 46% reduce 0%
2026-02-02 09:17:45,634 INFO mapreduce.Job: map 53% reduce 0%
2026-02-02 09:17:50,015 INFO mapreduce.Job: map 57% reduce 0%
2026-02-02 09:17:52,935 INFO mapreduce.Job: map 64% reduce 0%
2026-02-02 09:17:56,567 INFO mapreduce.Job: map 80% reduce 0%
2026-02-02 09:17:57,585 INFO mapreduce.Job: map 84% reduce 0%
2026-02-02 09:18:03,798 INFO mapreduce.Job: map 100% reduce 0%
2026-02-02 09:18:17,229 INFO mapreduce.Job: map 100% reduce 79%
2026-02-02 09:18:23,287 INFO mapreduce.Job: map 100% reduce 96%
2026-02-02 09:18:24,315 INFO mapreduce.Job: map 100% reduce 100%
2026-02-02 09:18:25,335 INFO mapreduce.Job: Job job_1770036565126_0002 completed successfully
2026-02-02 09:18:25,418 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=362926512
  FILE: Number of bytes written=545224850
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=264079713
  HDFS: Number of bytes written=29158010
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
```

Comprobamos el resultado

```
(kali@nodo1)-[~/Desktop/practica8]
$ /home/kali/Desktop/hadoop/bin/hdfs dfs -ls /resultado8
Found 2 items
-rw-r--r--  2 kali supergroup          0 2026-02-02 09:18 /resultado8/_SUCCESS
-rw-r--r--  2 kali supergroup 29158010 2026-02-02 09:18 /resultado8/part-00000
```

Vemos la salida

```
(kali@nodo1)-[~/Desktop/practica8]
$ /home/kali/Desktop/hadoop/bin/hdfs dfs -cat /resultado8/part-00000 | head

"CITED"
1
10000
100000
1000006
1000007
1000011
1000017
1000026
1000033
cat: Unable to write to output stream.

(kali@nodo1)-[~/Desktop/practica8]
$ █
```

4. Map Only

Ejecutamos Map Only

```
(kali@nodo1)~[~/Desktop/practica8]
$ /home/kali/Desktop/hadoop/bin/hadoop jar \
/home/kali/Desktop/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
-D mapred.reduce.tasks=0 \
-input /practicas/cite75_99.txt \
-output /resultado9 \
-mapper 'wc -l'

packageJobJar: [/tmp/hadoop-unjar2246687478344830241/] [] /tmp/streamjob2631555892621530226.jar tmpDir=null
2026-02-02 09:21:03,755 INFO client.DefaultNoHARMFaloverProxyProvider: Connecting to ResourceManager at nodo1/192.168.1.170:8032
2026-02-02 09:21:03,952 INFO client.DefaultNoHARMFaloverProxyProvider: Connecting to ResourceManager at nodo1/192.168.1.170:8032
2026-02-02 09:21:04,286 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/kali/.staging/job_1770036565126_0003
2026-02-02 09:21:04,731 INFO mapred.FileInputFormat: Total input files to process : 1
2026-02-02 09:21:04,759 INFO net.NetworkTopology: Adding a new node: /default-rack/192.168.1.170:9866
2026-02-02 09:21:04,760 INFO net.NetworkTopology: Adding a new node: /default-rack/192.168.1.171:9866
2026-02-02 09:21:04,951 INFO mapreduce.JobSubmitter: number of splits:2
2026-02-02 09:21:05,030 INFO Configuration.deprecation: mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-02-02 09:21:05,727 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1770036565126_0003
2026-02-02 09:21:05,728 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-02-02 09:21:05,891 INFO conf.Configuration: resource-types.xml not found
2026-02-02 09:21:05,898 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-02-02 09:21:05,997 INFO impl.YarnClientImpl: Submitted application application_1770036565126_0003
2026-02-02 09:21:06,059 INFO mapreduce.Job: The url to track the job: http://nodo1:8088/proxy/application_1770036565126_0003/
2026-02-02 09:21:06,062 INFO mapreduce.Job: Running job: job_1770036565126_0003
2026-02-02 09:21:16,632 INFO mapreduce.Job: Job job_1770036565126_0003 running in uber mode : false
2026-02-02 09:21:16,633 INFO mapreduce.Job: map 0% reduce 0%
2026-02-02 09:21:30,135 INFO mapreduce.Job: map 50% reduce 0%
2026-02-02 09:21:31,154 INFO mapreduce.Job: map 100% reduce 0%
2026-02-02 09:21:32,170 INFO mapreduce.Job: Job job_1770036565126_0003 completed successfully
2026-02-02 09:21:32,257 INFO mapreduce.Job: Counters: 33
  File System Counters
    FILE: Number of bytes read=0
    FILE: Number of bytes written=555978
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=264079713
    HDFS: Number of bytes written=18
    HDFS: Number of read operations=14
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=4
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=2
    Data-local map tasks=2
    Total time spent by all maps in occupied slots (ms)=21959
    Total time spent by all reduces in occupied slots (ms)=0
```

Comprobamos la salida

```
(kali@nodo1)~[~/Desktop/practica8]
$ /home/kali/Desktop/hadoop/bin/hdfs dfs -ls /resultado9
Found 3 items
-rw-r--r--  2 kali supergroup      0 2026-02-02 09:21 /resultado9/_SUCCESS
-rw-r--r--  2 kali supergroup    9 2026-02-02 09:21 /resultado9/part-00000
-rw-r--r--  2 kali supergroup    9 2026-02-02 09:21 /resultado9/part-00001

(kali@nodo1)~[~/Desktop/practica8]
$ s$
```

5. Creamos el script rand.py

Creamos el escript rand.py y le damos permisos de ejecución

```
(kali@nodo1)~[~/Desktop/practica8]
$ nano rand.py

(kali@nodo1)~[~/Desktop/practica8]
$ chmod +x rand.py

(kali@nodo1)~[~/Desktop/practica8]
$ ls -l rand.py
```

Ejecutamos con Streaming

```
(kali@nodo1)~[/Desktop/practica8]
$ /home/kali/Desktop/hadoop/bin/hadoop jar /home/kali/Desktop/hadoop/share/hadoop/tools/lib/hadoop-streaming-*.jar \
-D mapred.job.reducers=1 \
-input /practicass/cite75_99.txt \
-output /resultado1 \
-mapper 'python rand.py 10' \
-file rand.py
2026-02-02 09:52:27,031 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [rand.py, /tmp/hadoop-unjar3568826217342292238/] [] /tmp/streamjob9044340516019558566.jar tmpDir=null
2026-02-02 09:52:27,902 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at nodo1/192.168.1.170:8032
2026-02-02 09:52:28,124 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at nodo1/192.168.1.170:8032
2026-02-02 09:52:28,469 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/kali/.staging/job_1770036565126_0005
2026-02-02 09:52:29,020 INFO mapred.FileInputFormat: Total input files to process : 1
2026-02-02 09:52:29,042 INFO net.NetworkTopology: Adding a new node: /default-rack/192.168.1.170:9866
2026-02-02 09:52:29,044 INFO net.NetworkTopology: Adding a new node: /default-rack/192.168.1.171:9866
2026-02-02 09:52:29,191 INFO mapreduce.JobSubmitter: number of splits:2
2026-02-02 09:52:30,015 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1770036565126_0005
2026-02-02 09:52:30,015 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-02-02 09:52:30,206 INFO conf.Configuration: resource-types.xml not found
2026-02-02 09:52:30,206 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-02-02 09:52:30,583 INFO impl.YarnClientImpl: Submitted application application_1770036565126_0005
2026-02-02 09:52:30,613 INFO mapreduce.Job: The url to track the job: http://nodo1:8088/proxy/application_1770036565126_0005/
2026-02-02 09:52:30,615 INFO mapreduce.Job: Running job: job_1770036565126_0005
2026-02-02 09:52:38,818 INFO mapreduce.Job: Job job_1770036565126_0005 running in uber mode : false
2026-02-02 09:52:38,819 INFO mapreduce.Job: map 0% reduce 0%
2026-02-02 09:52:38,819 INFO mapreduce.Job: map 30% reduce 0%
2026-02-02 09:53:01,317 INFO mapreduce.Job: map 83% reduce 0%
2026-02-02 09:53:02,321 INFO mapreduce.Job: map 100% reduce 0%
2026-02-02 09:53:10,514 INFO mapreduce.Job: map 100% reduce 100%
2026-02-02 09:53:10,579 INFO mapreduce.Job: Job job_1770036565126_0005 completed successfully
2026-02-02 09:53:10,672 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=31362776
  FILE: Number of bytes written=63562533
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=264079713
  HDFS: Number of bytes written=28058456
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=2
  Launched reduce tasks=1
  Data-local map tasks=2
  Total time spent by all maps in occupied slots (ms)=41393
  Total time spent by all reduces in occupied slots (ms)=6303
  Total time spent by all map tasks (ms)=41393
  Total time spent by all reduce tasks (ms)=6303
```

6. Optimización

Ejecutamos el comando utilizando paralelismo (2 reducers)

```
(kali@nodo1)~[/Desktop/practica8]
$ /home/kali/Desktop/hadoop/bin/hadoop jar /home/kali/Desktop/hadoop/share/hadoop/tools/lib/hadoop-streaming-*.jar \
-D mapred.job.reducers=2 \
-input /practicass/cite75_99.txt \
-output /resultado_final_ejercicio \
-mapper 'python rand.py 10' \
-file rand.py
2026-02-02 09:57:49,433 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [rand.py, /tmp/hadoop-unjar17831152880869093680/] [] /tmp/streamjob737872051434359157.jar tmpDir=null
2026-02-02 09:57:50,612 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at nodo1/192.168.1.170:8032
2026-02-02 09:57:50,846 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at nodo1/192.168.1.170:8032
2026-02-02 09:57:51,200 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/kali/.staging/job_1770036565126_0006
2026-02-02 09:57:51,737 INFO mapred.FileInputFormat: Total input files to process : 1
2026-02-02 09:57:51,756 INFO net.NetworkTopology: Adding a new node: /default-rack/192.168.1.170:9866
2026-02-02 09:57:51,756 INFO net.NetworkTopology: Adding a new node: /default-rack/192.168.1.171:9866
2026-02-02 09:57:51,942 INFO mapreduce.JobSubmitter: number of splits:2
2026-02-02 09:57:52,240 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1770036565126_0006
2026-02-02 09:57:52,240 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-02-02 09:57:52,449 INFO conf.Configuration: resource-types.xml not found
2026-02-02 09:57:52,449 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-02-02 09:57:52,559 INFO impl.YarnClientImpl: Submitted application application_1770036565126_0006
2026-02-02 09:57:52,597 INFO mapreduce.Job: The url to track the job: http://nodo1:8088/proxy/application_1770036565126_0006/
2026-02-02 09:57:52,601 INFO mapreduce.Job: Running job: job_1770036565126_0006
2026-02-02 09:58:00,164 INFO mapreduce.Job: Job job_1770036565126_0006 running in uber mode : false
2026-02-02 09:58:00,165 INFO mapreduce.Job: map 0% reduce 0%
2026-02-02 09:58:21,270 INFO mapreduce.Job: map 23% reduce 0%
2026-02-02 09:58:23,433 INFO mapreduce.Job: map 46% reduce 0%
2026-02-02 09:58:28,623 INFO mapreduce.Job: map 57% reduce 0%
2026-02-02 09:58:29,685 INFO mapreduce.Job: map 73% reduce 0%
2026-02-02 09:58:30,716 INFO mapreduce.Job: map 83% reduce 0%
2026-02-02 09:58:31,724 INFO mapreduce.Job: map 100% reduce 0%
2026-02-02 09:58:42,123 INFO mapreduce.Job: map 100% reduce 100%
2026-02-02 09:58:44,228 INFO mapreduce.Job: Job job_1770036565126_0006 completed successfully
2026-02-02 09:58:44,316 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=31333580
  FILE: Number of bytes written=63504180
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=264079713
  HDFS: Number of bytes written=28032328
```

Comprobamos que ha funcionado

```
(kali@nodo1)~[/Desktop/practica8]
$ /home/kali/Desktop/hadoop/bin/hdfs dfs -ls /resultado_final_ejercicio
Found 2 items
-rw-r--r--  2 kali supergroup          0 2026-02-02 09:58 /resultado_final_ejercicio/_SUCCESS
-rw-r--r--  2 kali supergroup 28032328 2026-02-02 09:58 /resultado_final_ejercicio/part-00000
```